

The Compensation Paradox: A PLS-SEM Approach

Consulting Project

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```
# Load the dataset
df <- read_excel("../data/3_Data.xlsx")

head(df)
```

```
## # A tibble: 6 × 57
##       ID Q1_County Q1_Township Q1_Village Q3_InNP Q4_Gender Q5_Age Q6_Educatio
##   <dbl> <chr>      <chr>      <chr>      <dbl>    <dbl> <dbl>    <dbl>
## 1     1 Baoxing   Wulong     Tiepingshan  0        1     60
## 2     3 Baoxing   Wulong     Tiepingshan  0        1     57
## 3     4 Baoxing   Wulong     Tiepingshan  0        1     60
## 4     5 Baoxing   Wulong     Tiepingshan  0        0     49
## 5     6 Baoxing   Wulong     Dongsheng    1        1     67
## 6     7 Baoxing   Muping     Xueshan      0        0     56
## # i 49 more variables: Q7_HouseholdPop <dbl>, Q8_AnnualIncome <dbl>,
## #   Q9_AgriProportion <dbl>, Q10_MaizeArea <dbl>, Q10_PotatoArea <dbl>,
## #   Q10_TeeArea <dbl>, Q10_FruitArea <dbl>, Q10_HerbArea <dbl>,
## #   Q10_VegArea <chr>, Q10_RapeArea <dbl>, Q10_SweetPotatoArea <dbl>,
## #   Q11_TotalArea <dbl>, Q12_Cattle <dbl>, Q12_Sheep <dbl>, Q12_Pig <dbl>,
## #   Q12_Chicken <dbl>, Q12_Bee <dbl>, Q13_TotalLivestock <dbl>,
## #   Q14_CropAsset <dbl>, Q14_LivestockAsset <dbl>, Q14_HouseholdAsset <dbl>, ...
```

```
# Standardize the large financial variable to prevent convergence issues
df$TC2_EconomicLoss_Std <- scale(df$TC2_EconomicLoss)
```

2. Measurement Model Specification (Outer Model) We specify our measurement model here. Crucially, we model Intangible Costs and Policy Support as formative constructs (composite,

mode_B). This solves the methodological challenge from our initial CB-SEM attempt, allowing us to retain all nuanced survey items because they cause the latent cost, rather than reflect it.

```
# Define the measurement model
mm <- constructs(
  # Reflective construct (Indicators share the same underlying theme)
  reflective("Tolerance", multi_items("Tol", c("1_5", "2_1", "4_5"))),

  # Formative constructs (Indicators build the construct)
  composite("Intangible_Cost", multi_items("IC", c("1", "2_1", "2_2", "2_3")), weights = mode_B),
  composite("Policy_Support", multi_items("PS", c("1", "2")), weights = mode_B),

  # Single-item constructs
  composite("Intangible_Benefit", single_item("IB")),
  composite("Tangible_Cost", single_item("TC2_EconomicLoss_Std")),
  composite("Compensation", single_item("CE0"))
)
```

3. Structural Model Specification (Inner Model) Here we map Mengxi Kou's socio-ecological theory. We test the direct effect of compensation, and its indirect effects mediated through governance fatigue (Policy Support) and psychological burden (Intangible Costs).

```
# Define the structural paths
sm <- relationships(
  # Does compensation experience drive the mediators or tolerance directly?
  paths(from = "Compensation", to = c("Intangible_Cost", "Policy_Support", "Tolerance")),

  # Do the mediators and costs/benefits drive Tolerance?
  paths(from = c("Intangible_Cost", "Policy_Support", "Intangible_Benefit", "Tangible_Cost"),
        to = "Tolerance")
)
```

4. Model Estimation & Evaluation We estimate the model using the standard PLS algorithm. We will check the reliability of our reflective construct (Tolerance) and the discriminant validity (HTMT) to ensure our constructs are truly distinct.

```
# Estimate the model
pls_model <- estimate_pls(data = df,
                        measurement_model = mm,
                        structural_model = sm)

# Summarize the model
summary_pls <- summary(pls_model)

# 1. Check Reliability of Tolerance (Target: rhoA > 0.7, AVE > 0.5)
print("--- Reliability & Validity (Reflective Constructs) ---")
```

```
## [1] "--- Reliability & Validity (Reflective Constructs) ---"
```

```
summary_pls$reliability
```

```
##
##      alpha  rhoA  rhoC  AVE
## Compensation  1.000 1.000 1.000 1.000
## Intangible_Cost 0.280 1.000 0.366 0.263
## Policy_Support 0.183 1.000 0.500 0.496
## Intangible_Benefit 1.000 1.000 1.000 1.000
## Tangible_Cost 1.000 1.000 1.000 1.000
## Tolerance 0.577 0.613 0.599 0.337
##
## Alpha, rhoA, and rhoC should exceed 0.7 while AVE should exceed 0.5
```

```
# 2. Check Discriminant Validity (Target: HTMT < 0.90)
print("--- Discriminant Validity (HTMT) ---")
```

```
## [1] "--- Discriminant Validity (HTMT) ---"
```

```
summary_pls$validity$htmt
```

```
##
##      Compensation Intangible_Cost Policy_Support
## Compensation      .              .              .
## Intangible_Cost    0.193          .              .
## Policy_Support     0.539          1.112          .
## Intangible_Benefit 0.160          0.410          0.423
## Tangible_Cost      0.017          0.059          0.101
## Tolerance          0.225          0.755          0.372
##
##      Intangible_Benefit Tangible_Cost Tolerance
## Compensation      .              .              .
## Intangible_Cost    .              .              .
## Policy_Support     .              .              .
## Intangible_Benefit .              .              .
## Tangible_Cost      0.080          .              .
## Tolerance          0.336          0.082          .
```

5. Bootstrapping for Causal Significance (Mediation Analysis) To test our Research Questions rigorously, we use machine-learning-style resampling (Bootstrapping with 5,000 runs) to generate standard errors and confidence intervals for our path coefficients.

```
# Run the bootstrap (this may take a minute)
set.seed(123) # For reproducibility
boot_model <- bootstrap_model(seminr_model = pls_model, nboot = 5000)

summary_boot <- summary(boot_model)

print("--- Structural Path Significance ---")
```

```
## [1] "--- Structural Path Significance ---"
```

```
# If the 95% CI does not cross zero, the path is significant!
summary_boot$bootstrapped_paths
```

##		Original Est.	Bootstrap Mean	Bootstrap SD
## Compensation ->	Intangible_Cost	0.110	0.113	0.045
## Compensation ->	Policy_Support	-0.342	-0.344	0.030
## Compensation ->	Tolerance	-0.169	-0.168	0.049
## Intangible_Cost ->	Tolerance	-0.424	-0.432	0.052
## Policy_Support ->	Tolerance	-0.092	-0.091	0.060
## Intangible_Benefit ->	Tolerance	0.220	0.221	0.047
## Tangible_Cost ->	Tolerance	0.006	0.013	0.029

##		T Stat.	2.5% CI	97.5% CI	Bootstrap P Val
## Compensation ->	Intangible_Cost	2.457	0.022	0.198	0.017
## Compensation ->	Policy_Support	-11.578	-0.400	-0.285	0.000
## Compensation ->	Tolerance	-3.480	-0.263	-0.069	0.001
## Intangible_Cost ->	Tolerance	-8.150	-0.529	-0.329	0.000
## Policy_Support ->	Tolerance	-1.543	-0.212	0.023	0.129
## Intangible_Benefit ->	Tolerance	4.682	0.126	0.311	0.000
## Tangible_Cost ->	Tolerance	0.203	-0.030	0.087	0.705

```
# Calculate the Specific Indirect Effect (The Mediation)
print("--- Mediation: Compensation -> Intangible Cost -> Tolerance ---")
```

```
## [1] "--- Mediation: Compensation -> Intangible Cost -> Tolerance ---"
```

```
specific_effect_significance(boot_model,
                             from = "Compensation",
                             through = "Intangible_Cost",
                             to = "Tolerance",
                             alpha = 0.05)
```

```

##                                Original Est. Bootstrap Mean
## Compensation->Intangible_Cost->Tolerance      -0.047      -0.049
##                                Bootstrap SD T Stat. 2.5% CI 97.5% CI
## Compensation->Intangible_Cost->Tolerance      0.020  -2.373  -0.086  -0.009
##                                Bootstrap P Val
## Compensation->Intangible_Cost->Tolerance      0.017

```

6. Robustness: Multi-Group Analysis (MGA) To answer our final methodological challenge, we test if the theoretical framework behaves fundamentally differently for those who have engaged with the compensation system versus those who have not.

```

# # Split data based on Compensation Experience (CE0)
# # 0 = No Experience, 1 = Has Experience
# estimate_mga <- estimate_pls_mga(pls_model,
#                               condition = df$CE0 == 1,
#                               nboot = 5000)
#
# print("--- Multi-Group Analysis Results ---")
# # This tests if the difference in path coefficients between the two groups is statistically significant
# summary(estimate_mga)

# 1. Define a 'Structural' version of the model for MGA
# We remove Compensation from the measurement model because it's now our grouping factor
mm_mga <- constructs(
  reflective("Tolerance", multi_items("Tol", c("1_5", "2_1", "4_5"))),
  # Modeling these as composites (Formative) helps with the Low AVE/Alpha issue
  composite("Intangible_Cost", multi_items("IC", c("1", "2_1", "2_2", "2_3")), weights = mode_B),
  composite("Policy_Support", multi_items("PS", c("1", "2")), weights = mode_B),
  composite("Intangible_Benefit", single_item("IB")),
  composite("Tangible_Cost", single_item("TC2_EconomicLoss_Std"))
)

# 2. Update the Structural Model (Inner Model)
sm_mga <- relationships(
  paths(from = c("Intangible_Cost", "Policy_Support", "Intangible_Benefit", "Tangible_Cost"),
    to = "Tolerance")
)

# 3. Re-estimate the base model without the CE0 indicator
pls_model_mga <- estimate_pls(data = df,
                             measurement_model = mm_mga,
                             structural_model = sm_mga)

# 4. Run the MGA safely
# We use the CE0 column to define the groups
mga_results <- estimate_pls_mga(pls_model_mga,
                                condition = df$CE0 == 1,
                                nboot = 1000) # Start with 1000 to test, move to
                                              5000 for final

summary(mga_results)

```

```
##      source          target      estimate      group1_beta
## Length:4          Length:4      Min.       :-0.44288      Min.       :-0.36135
## Class :character   Class :character 1st Qu.: -0.15619      1st Qu.: -0.10280
## Mode  :character   Mode  :character Median    :-0.02597      Median    : 0.04267
##                                     Mean      :-0.06132      Mean      :-0.02397
##                                     3rd Qu.: 0.06889      3rd Qu.: 0.12150
##                                     Max.       : 0.24952      Max.       : 0.18014
##      group2_beta      diff      group1_beta_mean      group2_beta_mean
## Min.       :-0.49688      Min.       :-0.05323      Min.       :-0.412355      Min.       :-0.51044
## 1st Qu.: -0.20447      1st Qu.: 0.05447      1st Qu.: -0.191506      1st Qu.: -0.22162
## Median    :-0.06204      Median    : 0.10471      Median    :-0.006858      Median    :-0.07162
## Mean      :-0.09690      Mean      : 0.07293      Mean      :-0.059407      Mean      :-0.10565
## 3rd Qu.: 0.04553      3rd Qu.: 0.12316      3rd Qu.: 0.125240      3rd Qu.: 0.04435
## Max.       : 0.23337      Max.       : 0.13552      Max.       : 0.188441      Max.       : 0.23107
##      pls_mga_p
## Min.       :0.07247
## 1st Qu.:0.14875
## Median :0.35168
## Mean      :0.36179
## 3rd Qu.:0.56471
## Max.       :0.67132
```

Interpretation:

RQ1: Does compensation increase tolerance?

The Shocking Result: Compensation (CEO) has a direct negative effect on Tolerance ($\beta = -0.169$, $p = 0.001$). Simply put: People who apply for compensation actually end up less tolerant of wild boars.

The Indirect Path: Compensation also has a significant indirect negative effect through Intangible Costs ($\beta = -0.047$, $p = 0.017$). This means the policy adds a “psychological burden” or “governance fatigue” that further erodes tolerance.

RQ2: Tangible vs. Intangible Factors

Tangible Costs (Money): The effect of Tangible_Cost (Economic Loss) on Tolerance is not significant ($\beta = 0.006$, $p = 0.705$). This is a huge finding! It proves that the conflict is not about the money lost; it’s about how people feel.

Intangible Costs (Psychology): This is the strongest driver in the whole model ($\beta = -0.424$, $p = 0.000$). Emotional burden and perceived risk are what truly destroy tolerance.

Intangible Benefits (Values): Seeing the ecological or cultural value of the boars significantly boosts tolerance ($\beta = 0.220$, $p = 0.000$).

RQ3: Robustness & Multi-Group Analysis (MGA) Result: the MGA output shows that the minimum p-value for differences between groups (Applied vs. Not Applied) is 0.072.

Interpretation: Since no p-value is below 0.05, the structural model is robust. The psychological “rules” of how costs and benefits drive tolerance are the same for everyone, whether they have actually received a payout yet or not. This gives your results a high degree of generalisability.